

# MODELING CLIMATE CHANGE OVER TIME

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ABSTRACT. The goal of this paper is to examine historical climate change data and determine if we can find correlations and trends in CO<sub>2</sub> levels and changing temperatures. Most of our data examines trends in the United States, but we examine global trends as well. We conclude that the climate is changing rapidly and that the change is not due to volcanic activity, but that CO<sub>2</sub> abundance has a strong positive correlation to temperature.

## 1. PROBLEM STATEMENT AND MOTIVATION

It is well established that Earth’s atmospheric temperatures are seasonal. Aside from the four standard seasons associated with Earth’s axial tilt, long-range seasonal temperatures are influenced by ocean temperatures and myriad minute details of Earth’s orbit (called Milankovitch cycles) [Mil41]. While understanding these cycles is still an open area of investigation, they typically act in predictable ways over long periods of time. Within the last 200 years, however, Earth’s climate has changed faster than in any time known since the mass extinction event of the dinosaurs [Mul13].

With the climate changes on Earth trending toward another potential mass extinction event, there is great interest in identifying which factors have caused such a rapid change in global climate. Recently, there has been increasing momentum in both scientific and societal circles suggesting that Earth’s rapid shift in temperature is largely due to a rise in atmospheric CO<sub>2</sub> levels [Foo56, Arr96, Cal38, Kee60, MW67, CAB<sup>+</sup>79, STW<sup>+</sup>96]. Abstracted from the various factors lies the core question: how scared should we be of the trends indicated by the changes we see in the climate? We therefore attempt analysis and predictive modeling of these factors together over time, beginning as far back as data permits, until the present, and use these models to predict future behavior.

## 2. DATA

We have gathered data from various sources to analyze global trends and predict future temperatures. For average temperature data recorded

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throughout the globe, we have used data compiled by Berkeley Earth provided for free on Kaggle [Ber24]. This dataset contains average temperatures recorded in various countries with both land and ocean measurements recorded monthly. This dataset dates back to 1750, but some earlier years/months were not recorded. Thus, we primarily examine the average land temperature measurements that date back to 1753 in our analysis to avoid issues with missing data. We note that in earlier recorded years, there is much more variation in the recorded average temperature. This may be due to the limited number of locations reporting temperature data.

Another dataset contained averaged yearly temperatures from various points in the mainland United States from 1900 to 2023 [Ngu23]. This dataset required no cleaning.

To grasp the impacts of CO<sub>2</sub> and all greenhouse gas emissions on temperature, we use the emissions data provided by [RRR24] that contains most emissions caused by humans over the last 250+ years. The first iteration of this dataset that we used came from Kaggle, but after comparing the given values to their claimed source, we realized the Kaggle dataset had changed values that made United States pollution look worse and Chinese pollution look better than they truly are.

To compare this production to the CO<sub>2</sub> emissions caused by volcanoes, we consulted data collected by various experts in the area [FAC<sup>+</sup>19]. However, the amount of CO<sub>2</sub> released by volcanoes, while considerable, was less than 1% of the CO<sub>2</sub> that humans produced in the same time interval. As such, we did not consider volcano CO<sub>2</sub> production in our analysis.

We are considering time series data, and wish to predict future data. In this context, train-test splits without randomization may be helpful for model infrastructure selection. However, as the most recent data is likely to be most informative, we use all available data to train our final models displayed in the paper in the interest of predicting future states. All data, models, and figures may be found at our repository hosted via GitHub [Pri26].

### 3. METHODS

Every model draws assumptions as to the underlying behavior of the system itself. While some models may easily fit to existing climate data, some assumptions may be incompatible with the reality of climate change. This results in models generalizing badly to unseen future data. In the interest of seeking effective models for climate change, we consider several distinct model architectures operating under varying assumptions.

**3.1. Mixture Models.** One core assumption that many models hold is that all data was generated by the same system. Mixture Models avoid this assumption. Instead, they expect multiple models to be fit to different portions of the data. This allows for the possibility that the system was replaced or fundamentally altered somewhere in the training data, which

modified system will most influence data in the near future. The fundamental thesis of climate change is that the climate is directly and fundamentally changing, whether due to human impact or not. As such, it seems prudent to drop the assumption of homogeneous generation.

Gaussian Mixture Models are one form of Mixture Model. They operate by assuming that the data was generated by multiple multivariate Gaussian distributions, and assign membership likelihood of data according to the probability densities of the components at their locations. [Pea94] They are not often used on sequential, dependent time series, as another common assumption of their usage is that the data being fit are independent and identically distributed according to the fit variables. They are often biased to assume a split close to the center of the data temporally, as being a clear location for division of the data. However, they may still be used for modeling of time-series data, even incorporating time as a fit variable, leveraging assumptions of group membership and proximity similarity for identification of distinct processes generating the data. We will fit a Gaussian Mixture Model with two components, for simplicity assuming there is at most one fundamental shift, and then fit a linear regression to the data for each Gaussian component for explainability in the shifts of climate behavior.

**3.2. Kalman Filter.** The world is a large space, making it impossible to have complete observations of temperature, sea ice levels or CO<sub>2</sub> emissions globally. Moreover, new data streams in continuously as temperatures, seasons and even human activity fluctuate. This leads the Kalman filter to be a natural choice in modeling global average temperature trends.

As observations are limited, the Kalman filter’s natural smoothing of observations enables us to gain a much clearer understanding of the evolution of temperature trends over time. Moreover, the Kalman filter comfortably handles dynamic systems (like climate data) in updating its estimates and forecasts. The Kalman filter provides the statistically optimal estimate and is especially effective when dealing with Gaussian noise – which is a fair assumption to make for the noise of our global observations.

By pairing the Kalman filter with other methods, such as ARIMA (see Subsection 3.4), we may use exogenous data (e.g., CO<sub>2</sub> emissions) to make plausible forecasts for future global average measurements.

**3.3. Seasonal Time Series.** Another valuable tool for analyzing the overall temperature trend is a time series decomposition. This process involves decomposing a single observation series into three components: (1) a moving average of the data, (2) a seasonal variation (i.e., known temperature changes over the course of a calendar year) and (3) a remainder. To observe global long-term trends that extend across decades and centuries, we examine the world average temperatures over the last 200+ years [Ber24]. After computing a moving average, we compute the seasonal average with the detrended data. Finally, we compute the remainder to verify the accuracy of the seasonal trends identified.

**3.4. ARIMA Models.** ARIMA models use past time series data to forecast future predictions. It does so by using past values, past forecasting errors, and taking differences of consecutive terms to stabilize the mean of the time series. The general formula for an  $\text{ARIMA}(p, d, q)$  model is given by

$$\Delta^d Z_t = \mathbf{c} + \left( \sum_{i=1}^p \Phi_i \mathbf{y}_{t-i} \right) + \left( \sum_{j=0}^q \Theta_j \epsilon_{t-j} \right).$$

ARIMA models can be used for multivariate data, finding correlations between different features. For instance, we could examine correlations between temperature and  $\text{CO}_2$  levels, or between  $\text{CO}_2$  levels and ice levels.

ARIMA models are powerful because their coefficients are easy to interpret, yielding correlation between past measurements and current ones. Similarly, we can use ARIMA models to predict future states, enabling us to cast present results into the future.

These models, while powerful, are further improved by decomposing time series, as detailed in Subsection 3.3. Determining seasonal components and incorporating these into ARIMA models enable them to have more predictive power, since seasonal components do not interfere with general trends, or vice versa. These models, called SARIMA models, extend the ARIMA model framework.

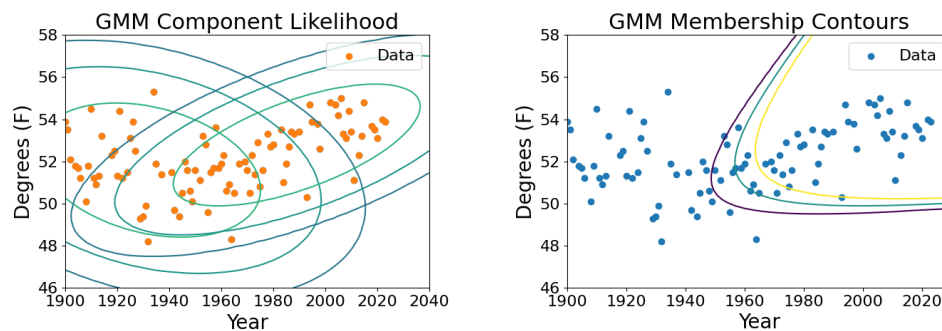
## 4. RESULTS AND ANALYSIS

**4.1. Mixture Models.** Consider the behavior in Figure 1a. From this figure, there is considerable overlap between the components, but the component to the right appears to indicate a distinct upward trend. This behavior is indicative of a shift in the underlying system generating the involved data. Figure 1b indicates that additional data upwards and to the right is more likely to have been produced by a new, fundamentally distinct system, with Figure 1c indicating that the present trend indicates a  $1F$  increase in US temperatures over each future 20-year interval.

**4.2. Kalman Filter.** Pictured in Figure 2 is a standard Kalman filter fit to all available temperature data from 1900 – 2017.

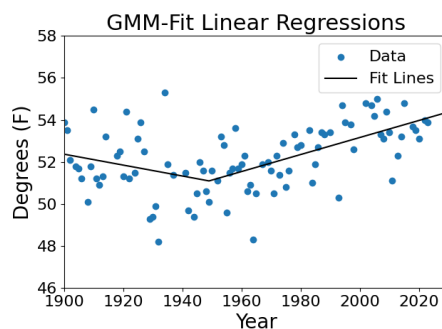
Leveraging the Kalman filter, we may apply a simple local linear model to predict future temperature trends. Pictured in Figure 3a, we train on US temperature data from 1900 – 1999 to forecast temperature from 2000 – 2017 and compare against later temperature observations.

However, what if we want to estimate the impact of conservation efforts? Once again, we may run the Kalman filter (via an ARIMA model, see Subsection 4.4) to predict the corresponding change in average temperature by annual reductions in human  $\text{CO}_2$  emissions. Figure 3b illustrates the change in average temperature corresponding to a 3% annual reduction in human  $\text{CO}_2$  emissions over the span of 50 years.



(A) Contour plot for the two-component Gaussian mixture model on average U.S. yearly temperatures, indicating the probability that any given data point was generated by that component. Note the earlier component (left), which exhibits a slight negative trend, and the modern component (right), which indicates a significant increase in temperatures over time.

(B) Membership probability bounds between the two components depicted in Figure 1a, showing 25%, 50%, and 75% bounds for the probability of membership in the more recent component.



(C) Linear regressions fit to each component shown in Figure 1a. Their intersection near 1950 suggests a shift from declining to increasing temperature trends, with an approximate increase of  $1^{\circ}F$  every twenty years after 1950.

FIGURE 1. Gaussian mixture model analysis of average U.S. yearly temperatures, including component contours, membership probability bounds, and fitted linear trends.

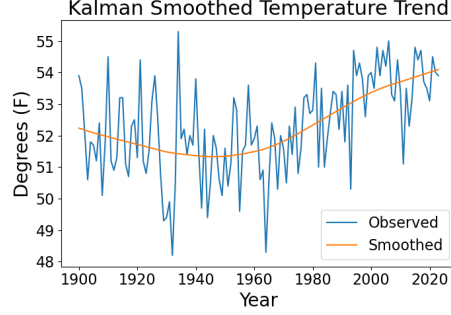
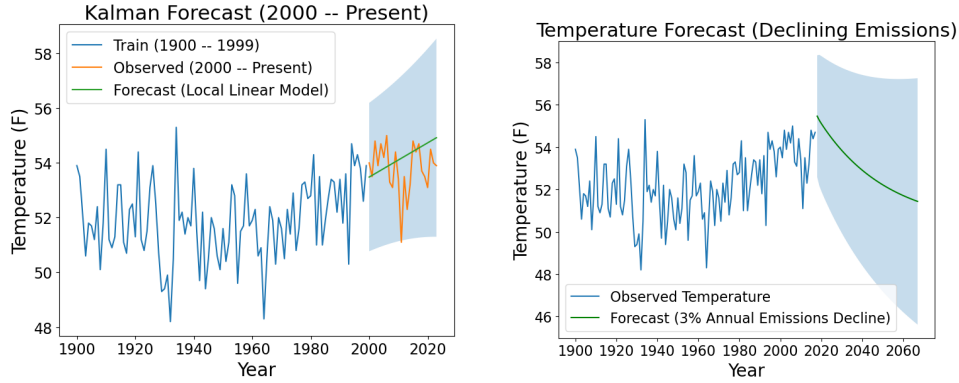


FIGURE 2. Average US land surface temperature data from 1900 – 2017. The Kalman filter, depicted in orange, smooths random year-over-year temperature noise to make climate trends readily apparent.

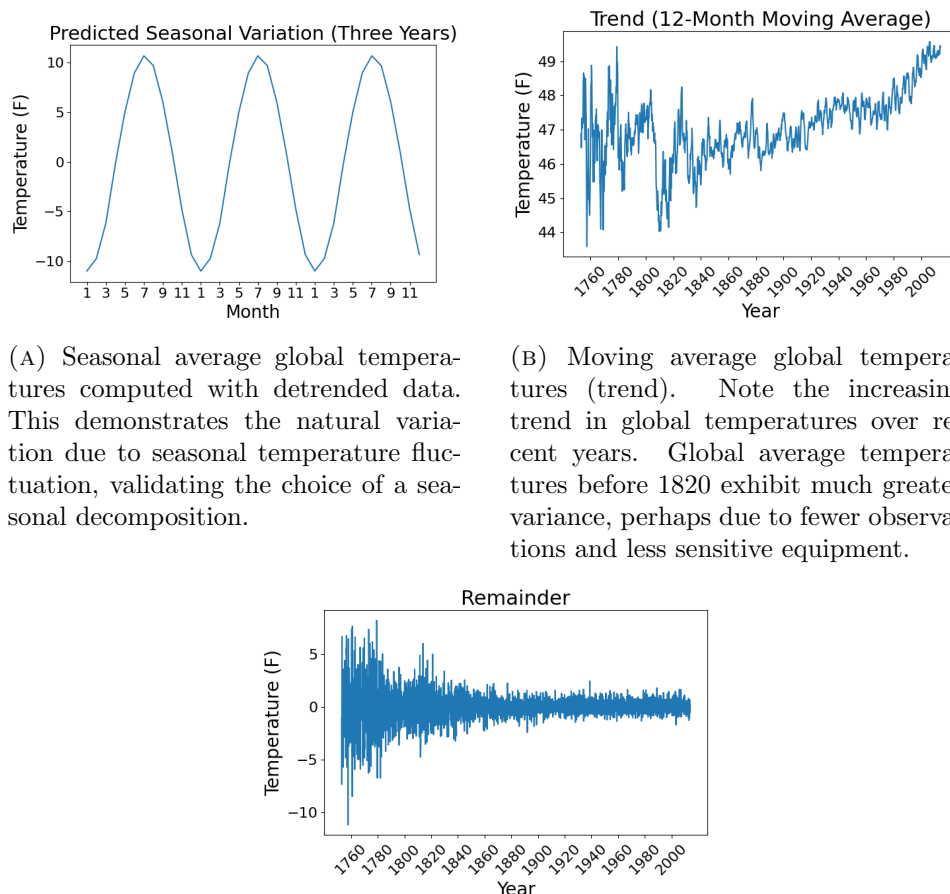


(A) Kalman filter forecast for average US temperature in years 2000–2017 compared against actual observed temperatures from the same period. The blue shaded region indicates the middle 95% confidence interval for the forecast.

(B) Predicted impact on US average temperature by reducing human  $\text{CO}_2$  emissions by 3% annually. After approximately 50 years, temperatures are expected to return to their 1900 level. The blue shaded region indicates the middle 95% confidence interval for the forecast.

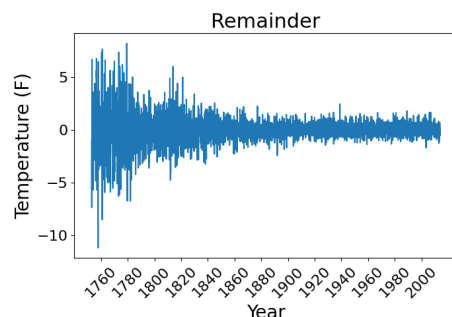
FIGURE 3. Kalman-based forecasting results. Depicted on the left is the increasing trajectory of US average temperature over recent years. Compare with the figure on the right, showing declining average temperatures conditioned on an annual human  $\text{CO}_2$  emission reduction of 3% annually.

**4.3. Seasonal Time Series.** We may observe that using our trend and seasonal components in Figures 4a and 4b, we can compute a remainder (shown in Figure 4c) that has a mean of approximately zero as the time



(A) Seasonal average global temperatures computed with detrended data. This demonstrates the natural variation due to seasonal temperature fluctuation, validating the choice of a seasonal decomposition.

(B) Moving average global temperatures (trend). Note the increasing trend in global temperatures over recent years. Global average temperatures before 1820 exhibit much greater variance, perhaps due to fewer observations and less sensitive equipment.



(C) Remainder temperatures (data minus trend minus seasonal). Note the mean is near zero with small variance.

FIGURE 4. Decomposition of the global average land temperature series from 1753–2017 into seasonal, trend, and remainder components.

series continues through 2017. The inconsistency in the remainder in early years is likely due to the fact that fewer locations were recording temperature data for a given year and had less sensitive equipment than is presently available. With the increase in consistency and accurate data, it is clear that temperatures are in fact rising, as shown in Figure 4.

**4.4. ARIMA Model.** ARIMA models can take a long time to train, and as such it is better to train them on aggregated or averaged data, instead of daily data. Computing the optimal parameters  $p$ ,  $d$ , and  $q$  using a grid search to match an ARIMA model to Kalman filter smoothed U.S. average yearly temperature resulted in the green line found in Figure 5. We also

trained the model on only the first 80% of the shown time series data, and still our predictions matched the smoothed data very closely.

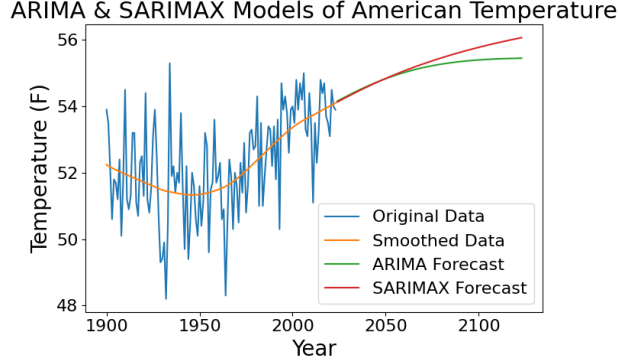


FIGURE 5. An ARIMA(2, 1, 2) model and related SARIMAX model fit to averaged US land temperatures from 1900-2024. The parameters were optimally selected from a grid-search. By incorporating long-term seasonal trends, the future prediction of US average land temperatures actually worsen.

Figure 5 clearly indicates that the temperatures in America have been increasing in recent decades, demonstrating a worrying trend.

An obvious question is whether or not natural multi-year cycles are taking place that cause this trend to appear more concerning than it may be in reality. As we cannot rule out decade-long trends that may skew ARIMA model predictions, we trained a SARIMAX model to determine seasonal trends in addition to the standard ARIMA techniques, resulting in an even more foreboding prediction: as indicated by the red line in Figure 5, temperatures are forecast to be even higher than ARIMA predicted.

## 5. ETHICAL IMPLICATIONS AND CONCLUSIONS

Few modern issues are as divisive and wide-reaching as that of human influence on the global climate. Consequently, it is essential for any investigator to carefully analyze their methods before making any definitive claims. While the collection of climate data is generally ethical and free of any harm, the ignorance of the story the data indicate is not.

With such a heated debate being had globally, and myriad conflicting interests, we emphasize that our results do not assign culpability to any individual person, industry or nation. Instead, the only proper interpretation of our results is for all people everywhere to recognize the global trends correlated to human CO<sub>2</sub> emissions and hold themselves individually responsible for whatever impact they may be able to make on their environment.

To avoid misuse and misunderstanding of the data, we emphasize that global climate trends are noisy. It may take decades to gain a clear picture of climate trends today. Thus, while daily weather events may dominate the headlines, we urge all readers to keep in mind that climate change is a long-term threat and requires longitudinal context to fully understand.

Similarly, we remind the reader that one of the datasets we found on Kaggle contained modified data that made the United States appear to be contributing more pollution than they are truly creating. The fact that we discovered one such incorrect dataset indicates that there are likely many more that we have not, which begs the question if this and other datasets may be being used for propaganda purposes.

In analyzing our results, we do not hope to single any individual or group out for their actions. Rather, we encourage all people everywhere to develop a greater awareness of the long-term climate trends and what we each may do individually to improve our shared environment.

## 6. CONCLUSION

From our combined analysis, we find that Earth’s average temperature has continuously and rapidly trended upward over the last 75 years. Our models indicate this trend will continue without significant intervention, particularly in human CO<sub>2</sub> emissions. Similarly, we found that volcanoes produced less than 1% of the amount of CO<sub>2</sub> that humans have in the time intervals that we examined, implying that they are likely not the cause of significant climate change.

According to our analysis, US average temperatures are forecast to reach 56 degrees Fahrenheit by 2100 at our current trajectory (see Figure 5) – marking a 4.5-degree increase from 1950. Contrast this with the Kalman forecast for declining CO<sub>2</sub> emissions (see Figure 3b). With just a 3% annual reduction in CO<sub>2</sub> emissions, we find US average temperatures are forecast to return below 52 degrees Fahrenheit (consistent with average temperatures during the early 20th-century) within a five-decade timespan.

In short, our analysis paints a clear picture: reducing carbon emissions may effectively reverse the unprecedented increase in global average temperatures. However, without these reduction efforts, average temperatures will continue to rise further and further beyond their historical baseline. The global climate outcomes of future decades are heavily governed by our emissions today, only further underscoring the critical importance we face today as we respond to nature’s call for help.

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